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FoodEx2 maintenance 2020

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Abstract

FoodEx2 is a comprehensive food classification and description system developed and maintained by EFSA with the aim to keep it up to date and to cover the needs of data collections in different domains. To ensure its full applicability, a regular maintenance of the system is needed. Three major updates were carried out so far and EFSA reported on the applied changes annually. This technical report describes the outcome of the fourth maintenance process done during 2020, which entailed addition of new terms, deprecation of terms and amendments to existing terms including enrichment of implicit facets and changes to the parent-child relationship.

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Summary

A standardised food classification and description system such as FoodEx2 is necessary to provide a common link to data sources coming from different food safety domains. In order to meet this requirement, it needs to be actively maintained, remain up to date as well as flexible to accommodate data collections in different domains. To this end an annual maintenance of the system has been taking place since April 2015 when a major revision (FoodEx2 revision 2) was performed.

A maintenance technical group composed of EFSA staff with expertise in different domains (food consumption, chemical contaminants, pesticides residues, etc.) was set-up to evaluate proposals and requests provided by users and stakeholders, take decisions regarding their implementation and inform them on the outcome.

During the maintenance process performed in 2020 the following changes have been implemented: the scope of some existing terms is better specified or expanded by changing the name, scientific name, adding alias or other implicit attributes; type of term is changed for six codes and level of detail for 10 codes; 80 terms are dismissed¹ in some of the hierarchies (six due to the alignment with the Commission Regulation (EU) 609/2013 and 74 due to redundancy); three terms are deprecated² due to duplication and 37 due to redundancy.

Amendments in different sections of the catalogue led to the addition of 108 new terms in different hierarchies and facets: Honey bees section is amended with 13 species and subspecies as needed for data collection within the MUST-B project; Birds species are amended with 68 new terms as needed for Avian Influenza Surveillance programme and completion of taxonomy for existing species; the facet F31-Animal-age-class is amended with nine new terms for the SIGMA project purposes, three terms are added the facet F20-Part-consumed-analysed to accommodate precise coding of the scallops samples within the Marine-biotoxin data collection; four terms are added to the Botanical hierarchy to accommodate maintenance of the EFSA Compendium of Botanicals database; four terms are added as need for the EFSA's Chemical Hazard Database (OpenFoodTox); seven terms are added to different hierarchies and facets to accommodate various requests from internal or external users.

The applicability of some terms is extended to further hierarchies and/or facets and the position or type of some terms in the hierarchy tree is changed since a more appropriate parent-child term relationship is found; implicit facets are updated for some terms to keep consistency with the logic of the FoodEx2 system.

The present report describes the outcome of the fourth maintenance process performed in 2020 in order to keep track of the changes in the system within two minor releases published in 2020 (MTX 11.1 and MTX 11.2) and the major release published in January 2021 (MTX 12.0).

¹ Dismissed terms are made not reportable in one or more hierarchies of the FoodEx2 catalogue, but they continue to be reportable in others

² Deprecated terms are made not reportable in all hierarchies of the FoodEx2 catalogue

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1. Introduction

1.1. Background and Terms of Reference as provided by the requestor

In 2011, EFSA released a comprehensive food classification and description system named FoodEx2 (revision 1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision 2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011b, 2015), internal and external users should provide to EFSA proposals for implementation and maintenance in order to refine the system. In fact, innovation in the food market and gained experience by using the system led to the need of keeping the catalogue updated. Following the revision 2 of the system, the FoodEx2 catalogue was revised three times based on EFSA needs and evaluation of the data providers and stakeholders' suggestions. A first maintenance took place in 2015 (EFSA et al., 2016) followed by another extensive one in 2016-2018 and a third in 2019. The updated catalogues (MTX 10.0 and MTX 11.0) and the technical reports that summarize all changes performed during these maintenances are published (EFSA et al, 2019a, 2020).

Moreover, harmonized data collection in the areas of pesticides, contaminant occurrence, veterinary medicinal product residues and food additives (EFSA, 2021a) is reflected in the maintenance process of the FoodEx2 catalogue. Considering that all the above data domains and number of other data collections in EFSA use the FoodEx2 catalogue, there was a need to accommodate different requests. Some changes were performed to enable giving a clear guidance (in terms of FoodEx2 codes) to data providers before the data collection opening date, while others were requested from EFSA staff working in different domains, data providers and network members (hereafter referred to as users).

By using the system, both internal and external users and stakeholders, provided constructive comments and requests for the addition of missing terms and other type of updates in different sections of the catalogue. All these suggestions were considered for the maintenance purposes. A technical group (TG) composed of EFSA staff with expertise in different domains carried out the maintenance process evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the fourth maintenance process done in 2020 in order to keep track of the changes in the system done within two minor releases published in 2020 (MTX 11.1 and MTX 11.2) and the major release published in January 2021 (MTX 12.0). In order to perform a correct coding, with a reasonable level of detail, this report should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described.

2. Data and Methodologies

2.1. Data

The FoodEx2 catalogue MTX 11.0 - version published in January 2020 was used as a starting point. The suggestions provided during the period between January 2020 and January 2021 by users and stakeholders of the FoodEx2 system were collected through written feedback and used to update the system.

2.2. Methodologies

Each request for addition or changes in the structure was considered and, based on expert judgment, the TG decided whether a change was needed, or the request was already covered by the available terms and facet descriptors. All the changes were made following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1. Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Catalogue Browser³ (EFSA and Ioannidou, 2019), a user-friendly tool developed in Java by EFSA staff (Evidence Management Unit) to help create FoodEx2 codes and to facilitate the management of the MTX catalogue.

2.2.2. Maintenance process

In order to carry out the fourth maintenance of the FoodEx2 catalogue, the TG performed the following steps:

- Evaluated all requests, comments and proposals from internal and external users and stakeholders
- Decided on their implementation based on the following options: add new terms, extend or better specify the scope of existing terms, identify the correct way of implementing the suggestion using already existing terms and rules, change the parent-child relationship of the terms, deprecate terms and set terms as not reportable/dismissed in one hierarchy but still reportable in others
- Checked for inconsistencies and ambiguities
- Updated the FoodEx2 catalogue
- Summarised the changes for the present technical report.

3. Maintenance output

3.1. Addition of new terms

New terms added to the catalogue in 2020 maintenance were in total 108. Most of them (n = 85) are added as natural sources only to the MTX hierarchy and F01 – Source facet. The bird section gained 68 terms (see section 3.4), the section related to Honey bees was amended with 13 new terms (see section 3.5) while four new terms were added in the section on Crustaceans, requested for the maintenance of the EFSA's Chemical Hazard Database (OpenFoodTox)⁴. In order to accommodate the maintenance of the EFSA Compendium of Botanicals database⁵, four new terms were added to the MTX and Botanical hierarchies. To ensure comprehensive and precise coding of the scallop samples (collected within the Marine-biotoxins data collection and needed for the EFSA mandate on Lipophilic toxins and Domoic Acid in scallops⁶) three new terms were added to the F20 – Part-consumed-analysed facet. FoodEx2 is also used for mapping the samples within the SIGMA animal disease data model (EFSA 2021a). In order to classify samples by generic age classes, applicable to all animals, five terms were added to the F31 – Animal-age-class facet. In addition, to accommodate description of the animal samples with respect to the post-mortem intervals, four new terms were added to the same facet (see section 3.6). An additional seven terms were added in different hierarchies and/or facets requested by internal or external users for different purposes.

During the maintenance in 2020, 126 requests for new terms were received by internal and external users of the catalogue. About 65% of them were not considered for addition for the reasons listed below:

³ <https://www.efsa.europa.eu/en/data/data-standardisation>

⁴ <https://www.efsa.europa.eu/en/data/chemical-hazards-data>

⁵ <https://www.efsa.europa.eu/en/data/compendium-botanicals>

⁶ M-2020-0052; EFSA-Q-2020-00245- Request for a scientific opinion on the shucking of certain species of scallops contaminated with lipophilic toxins with a view to the production of edible parts meeting the safety requirements foreseen in the Union legislation; EFSA-Q-2020-00246 - Request for a scientific opinion on the shucking of certain species of scallops contaminated with domoic acid with a view to the production of edible parts meeting the safety requirements foreseen in the Union legislation

- Terms were already present in the catalogue either as a base term or combination of a base term and facet descriptors according to the logic of the system (EFSA, 2015). For all these requests the TG provided the coding solutions to the users
- Request was not in line with the FoodEx2 logic and/or relevant to EFSA's data collections.

A detailed table with all additions performed during 2020, the hierarchies and facets affected and the rationale behind is available in Appendix A –.

3.2. Changes in existing terms

Taking into account all suggestions from users and keeping the consistency of the system, amendments to existing terms were applied. Term extended names were updated for 81 codes, common names were added or updated for 95 terms while scientific name/synonyms were added or updated for 97 terms. These changes were: 1) editorial, such as singular/plural form or correction of misspelling, 2) update of the term extended name, update/addition of new common/scientific names and removal of existing ones to ensure more accurate description of the term. Consequently, the scope notes were updated for 193 terms. The majority of these changes were done within the revision of the Birds section (see section 3.4).

In order to improve the hierarchical structure of the catalogue and adhere to the rules for the correct use of the system, the level of detail was changed for 10 terms as presented in Table 1. In addition, for the last six terms in Table 1, for which the level of detail was changed from 'Core term' to 'Facet descriptor', the type of term was also changed from 'Broad or mixed' to 'Facet' (see section 3.8), since they were set as not reportable in all FoodEx2 hierarchies, but remained as descriptors under the facet F02 - Part-nature.

Table 1: Updates on level of detail during the maintenance in 2020

CODE	NAME	Old level of detail	New level of detail
A00KS	Lettuces and salad plants	Hierarchy term	Non-specific term
A04MN	Fruit / vegetable spreads and similar	Hierarchy term	Aggregation term
A04TG	Honey bees (as animal)	Non-specific term	Core term
A04TH	Melipona (as animal)	Extended term	Core term
A0C6Z	Air	Core term	Facet
A0C70	Delivery box liner	Core term	Facet
A0C71	Dust	Core term	Facet
A0C72	Fabric swab	Core term	Facet
A0C73	Hatcher basket liner	Core term	Facet
A0C74	Boot swabs	Core term	Facet

Matrix code was changed for two and removed from one FoodEx2 code in accordance with the Commission Regulation (EC) 396/2005⁷. For the term A01EB - Wild strawberries, the old matrix code 'P0152010-002' was changed to 'P0152000-002' and for the term A0DEE - European bamboo, the old matrix code '0270080-001' was changed to 'P0270080-001'. In addition, the matrix code 'P1300000' was removed from the term A0CTR - Processed food products, as it is not listed in Annex I to Regulation (EC) No 396/2005⁸. Finally, the matrix code 'P0840020' was attributed to the newly added term A18FY - Ginger roots spice in accordance with the same Annex to Regulation (EC) No 396/2005 (Appendix A –).

⁷ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32005R0396&from=EN>

⁸ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018R0062&qid=1612880652435&from=EN>

3.3. Amendments to implicit facets

3.3.1. Amendments in Feed area

According to the logic of the FoodEx2 system, Raw Primary Commodity (RPC) terms are attributed with implicit facets F01 – Source and F27 – Source-commodities (EFSA, 2015). Similarly, all RPC derivatives have implicitly assigned an F27 – Source-commodity facet being the correspondent RPC term. So far, all feed related terms were attributed with implicit facets F02 – Part-nature and F23 – Target-consumer. In addition, the RPC terms in the Feed section were attributed with the F01 – Source facet and RPC derivatives were attributed with the facet F27 – Source-commodities. During the FoodEx2 maintenance in 2020, the F27 – Source-commodities facet was added, as implicit, to all RPC terms (n = 80) i.e. the terms themselves are attributed as implicit F27 – Source-commodities facet. These changes are not presented in detail in this report, but they are implemented in the catalogue and therefore present in Appendix B –.

3.3.2. Other amendments to implicit facets

Implicit facets were updated for seven terms available in different hierarchies and facets after being recognized as insufficient or incorrect by TG or data providers. Changes performed on implicit facets for these seven terms are presented in Table 2: .

Table 2: Updates on implicit facets on non-feed related terms during the maintenance in 2020

FoodEx2 code	Term Extended Name	Old implicit facets	New implicit facets
A01AD	Other Root and rhizome spices	F01.A0BXQ\$F02.A0F0F\$F27.A01AD	F01.A05VE\$F02.A0F0F\$F27.A01AD
A03QY	Simple cereals which have to be reconstituted with milk or other appropriate nutritious liquids	F02.A06FK\$F27.A000L\$F28.A07KG	F02.A06FK\$F28.A07KG\$F04.A0EZF
A02FD	Crustaceans	F01.A0BXN\$F02.A0EMF	F01.A0BXN\$F02.A0EMF\$F27.A02FD
A038L	Processed fat (blubber) of marine mammals	F01.A056K\$F02.A06AR\$F27.A0F3G\$F28.A0C02	F02.A06AR\$F27.A18FX\$F28.A0C02
A065H	Baked milk and similar	F02.A0DSP\$F27.A02LT\$F28.A07HJ	F02.A0DSP\$F04.A02LT\$F28.A07HJ
A0CAZ	Duck foie gras animal (as animal)	F01.A058B\$F29.A0C7G\$F31.A0C8P	F01.A058B\$F29.A18FV\$F31.A0C8P
A0CBG	Goose foie gras animal (as animal)	F01.A058C\$F29.A0C7G\$F31.A0C8P	F01.A058C\$F29.A18FV\$F31.A0C8P

\$ - separator between facet descriptors

3.4. FoodEx2 update in accordance with Avian Influenza Surveillance Programmes

FoodEx2 is used for reporting data under the Avian Influenza (AI) surveillance programmes in poultry and wild birds carried out by Member States according to the Commission Decision 2010/367/EU⁹ and

⁹ Official Journal of the European Union, Commission Decision of 25 June 2010 on the implementation by Member States of surveillance programmes for avian influenza in poultry and wild birds (notified under document C(2010) 4190) (Text with EEA relevance) (2010/367/EU)

guidelines provided by Council Directive 2005/94/EC¹⁰ (EFSA et al., 2018). Given that an extensive amendment of the Bird section (A057X – Bird (as animal)) was done in 2019 (EFSA et al., 2020), only seven new terms were requested for addition by data providers, as detailed in Appendix A. Nevertheless, this section required additional structuring in terms of completion of taxonomy and parent-child relationships. Moreover, considering continual changes of the taxonomy of wild bird species, an annual maintenance of the Birds section in FoodEx2 catalogue is needed. In order to support this maintenance, the Animal and Plant Health (ALPHA) Unit in EFSA engaged an external expert in ornithology, under a negotiated procedure, to perform a revision of the current content and propose necessary changes and amendments annually for the period of the next three years. This revision involves detailed evaluation of the term names, common names, scientific names/synonyms, scope notes, Euring codes, completion of taxonomy (missing terms for order, family and genus of existing species) and parent-child relationships. As a result of this activity, during the maintenance in 2020, term name, common name, scientific name/synonym and/or scope notes was updated for 191 terms and Euring code was added to one term (A0CTM - Emu (as animal)). In addition, 61 new terms were added to the Bird section in order to complete taxonomy for existing species (Appendix A –) which consequently led to the update of the parent terms for 148 codes.

3.5. Amendment of the Hymenoptera section in accordance with MUST-B project

FoodEx2 catalogue is used for data collection within the MUST-B project¹¹ ('EU efforts towards a holistic and integrated risk assessment approach of multiple stressors in bees'). During development of specification for the MUST-B field data collection, different honey bee subspecies in North, Central and South Europe are identified (EFSA, 2017; EFSA et al, 2021). To allow MUST-B project contractors to report data on these subspecies, the A0CMJ – Hymenoptera (as animal) group in the FoodEx2 catalogue was amended with 13 new codes as presented in Table 3: . In addition, the level of detail was changed for the term A04TG – Honey bees (as animal) from Non-specific term (yellow sphere) to Core term (red sphere). The term A04TH - Melipona (as animal) is moved one level up as a direct child of the A0CMJ – Hymenoptera (as animal) based on literature (EFSA et al, 2021; Engel, 1999) and consequently its level of detail changed from Extended term (green sphere) to Core term (red sphere).

Table 3: New terms added in the Hymenoptera group

FoodEx2 code	Term name	Scientific name	Parent term	Reportability
A18GK	European dark bee	<i>Apis mellifera mellifera</i>	A04TJ	MTX, F01 - Source
A18GL	Carnolian honey bee	<i>Apis mellifera carnica</i>	A04TJ	MTX, F01 - Source
A18GM	Iberian honey bee	<i>Apis mellifera iberiensis</i> \$ <i>Apis mellifera iberica</i>	A04TJ	MTX, F01 - Source
A18GN	Italian honey bee	<i>Apis mellifera ligustica</i>	A04TJ	MTX, F01 - Source
A18GP	Greek honey bee	<i>Apis mellifera cecropia</i>	A04TJ	MTX, F01 - Source
A18GQ	Cretan honey bee	<i>Apis mellifera adami</i>	A04TJ	MTX, F01 - Source
A18GR	Macedonian honey bee	<i>Apis mellifera macedonica</i>	A04TJ	MTX, F01 - Source
A18GS	Cyprian honey bee	<i>Apis mellifera cypria</i>	A04TJ	MTX, F01 - Source
A18GT	Maltese honey bee	<i>Apis mellifera ruttneri</i>	A04TJ	MTX, F01 - Source
A18GV	Asian honey bee	Eastern honey bee\$ <i>Apis cerana</i>	A04TG	MTX, F01 - Source
A18GX	Black Dwarf honey bee	<i>Apis andreniformis</i>	A04TG	MTX, F01 - Source
A18GY	Giants honey bee	<i>Apis dorsata</i>	A04TG	MTX, F01 - Source
A18GZ	Red dwarf honey bee	<i>Apis florea</i>	A04TG	MTX, F01 - Source

\$ - separator between two common names

¹⁰ COUNCIL DIRECTIVE 2005/94/EC of 20 December 2005 on Community measures for the control of avian influenza and repealing Directive 92/40/EEC

¹¹ <https://www.efsa.europa.eu/en/topics/topic/bee-health>

3.6. FoodEx2 updates in accordance to the SIGMA project requirements

FoodEx2 is used for mapping the samples collected under the SIGMA project – Animal Disease Data Model (σ -ADM) (EFSA et al, 2019b). To accommodate mapping of animal samples and indicate their Post-mortem interval, the facet F31-Animal-age-class was amended with a new branch of four terms. The term Post-mortem was added as a parent (root of the branch) together with its three children that allow classification of the fresh, decay and skeletonized stage of the animal's dead body. In addition, samples collected within the SIGMA project should be classified with general, i.e. not animal specific age classes. The previously available content of the F31 facet was not sufficient to cover the SIGMA project needs and therefore the TG was asked to add four new terms: Embryo/Unhatched egg, Juvenile - new born, Juvenile - immature, Adult. To accommodate this request, and to emphasize that these terms should be used only for the SIGMA project purposes, a new branch was added to the F31 facet, 'Age classes for SIGMA' as a parent to the four previously mentioned new terms and was set as not reportable i.e. only its children could be used for the SIGMA project purposes. The current view of the F31-Animal-age-class facet, with new branches expanded, including nine new terms in total, is presented in Figure 1 and Appendix A. Furthermore, the facet F29-Purpose-of-raising is enriched with one new term A18FV - Foie gras production purpose, as needed for the SIGMA project purposes (Appendix A –).

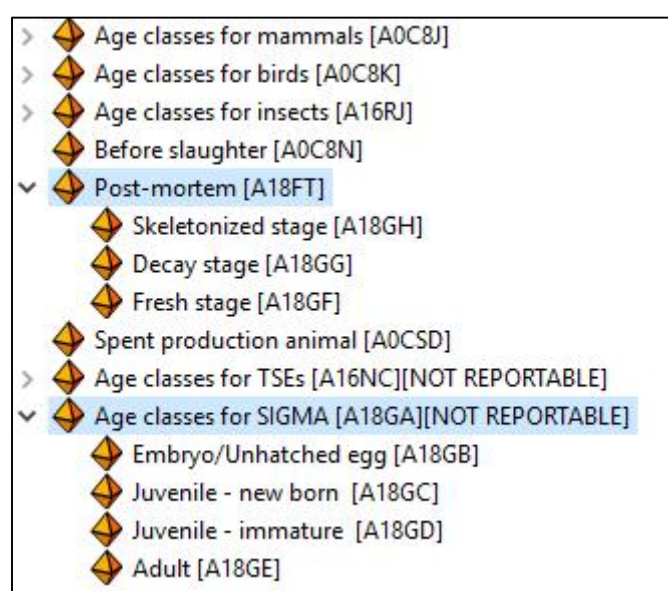


Figure 1: Current view of the F31-Animal-age-class facet – snapshot from the FoodEx2 catalogue browser

3.7. Update of the Exposure hierarchy in accordance to the Regulation (EU) 609/2013

FoodEx2 catalogue is regularly updated according to the EU Commission Regulations. Regulation (EU) 609/2013¹² has been consolidated. Particularly, the food category 'Food for sporting people' has been removed given that a report adopted by the European Commission on food intended for sports people concluded no necessity for specific provisions for foods intended for sports people which, from July 2016, are covered by horizontal rules of EU food law¹³. In order to align the FoodEx2 catalogue, this food category and its children were dismissed in the Exposure hierarchy and therefore are no longer available for reporting in the Exposure hierarchy as shown in Figure 2: . In the next major release of the FoodEx2 catalogue, that will take place in January 2022, they will be deprecated i.e. they won't be available in any hierarchy or facet.

¹² <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:02013R0609-20170711&qid=1591889565493&from=EN>

¹³ https://ec.europa.eu/food/safety/labelling_nutrition/special_groups_food_en

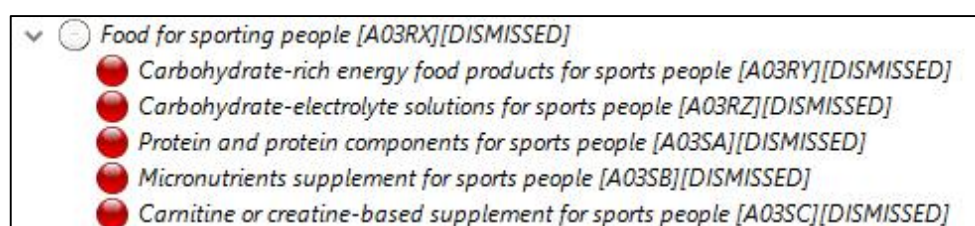


Figure 2: Section on Food for sporting people, snapshot from the FoodEx2 catalogue browser

For the future reporting of such foods with the FoodEx2 codes, terms that more precisely depict consumed/analysed food or drink should be used. For example, various energy bars (cereal bars and other sweet bars) previously reported under 'Carbohydrate-rich energy food products for sports people' should now be mapped to A00EY - Cereal bars or A0EQR - Sweet bars and other formed sweet masses. Similarly, energy or isotonic drinks previously reported as 'Carbohydrate-electrolyte solutions for sports people' should now be mapped to A03GA - Energy drinks and A03GB - Isotonic and sport drinks. Other codes could be considered in case of better matching with the food type/origin. Codes should be amended with facets if need. The codes A03SB - Micronutrients supplement for sports people and A03SC - Carnitine or creatine-based supplement for sports people should be mapped as any other food supplement i.e. starting from the section A03SJ - Food supplements and similar preparations, adding specific vitamins, chemicals or other components via F04 - Ingredient facet and specifying form via F03 - Physical-state facets. Finally, the code A03SA - Protein and protein components for sports people should be mapped to A02PN - Whey powder or similar foods in case of better matching.

3.8. Deprecated and dismissed terms during the maintenance 2020

During the regular update of the FoodEx2 catalogue, the TG identified 40 redundant terms that were deprecated and therefore not anymore available in the MTX 12.0 release. Two out of these 40 terms were related to the bird's species, previously available in the MTX hierarchy and the F01-Source facet only (A0CDX - Crows (as animal) and A0CHQ - Owls (as animal)). One term was related to the plant species, available only in the MTX and Botanicals hierarchies (A17FP - Moringa aptera Gaertn. (as plant)). Since the TG identified their duplicates whose description and implicit attributes fully cover the reporting needs, these three terms were deprecated. The rest of deprecated terms (n = 37) were previously present only in the MTX and Exposure hierarchy under the section A04KQ - Groups for exposure view. The whole section was carefully reviewed, and these terms were considered redundant, considering the logic of the FoodEx2 system. According to the rules of coding, to describe a term that is not listed under a certain category, a user should start from the parent term and describe it using the facets F01-Source, F27-Source-commodities or F04-Ingredient, if the term type is a RPC, RPC-derivative or Composite food respectively (EFSA, 2015). If the consumed/analysed food cannot be further specified with these three facets, the facet 'F26.A07XE - GENERIC TERM = Other' should be added. Moreover, children terms previously present under the codes: A04NV - Miscellaneous fresh uncured cheeses, A04PS - Other (mixed) fruit and vegetable juices or nectars, A04PP - Other fruit nectars, A04QH - Other liqueurs, A04QR - Other regulated additives, A04PR - Other vegetable juices are now moved one level up and therefore, their parent term was changed. All deprecated terms and the rationale behind are presented in Table 4: .

Table 4: List of deprecated terms in the FoodEx2 catalogue during the maintenance in 2020

FoodEx2 code	Term name	Reason for deprecation
A0CDX	Crows (as animal)	Overlapping with A17RN - Corvus (as animal)
A0CHQ	Owls (as animal)	Overlapping with A0CSX - Strigiformes (as animal)
A17FP	Moringa aptera Gaertn. (as plant)	Overlapping with A17FQ Moringa peregrina (Forssk.) Fiori (as plant)
A043E	Extracts	Overlapping with the term A047C - Essences and extracts
A04QJ	Seasonings and extracts	Overlapping with the term A04QK - Seasonings

FoodEx2 code	Term name	Reason for deprecation
A04NV	Miscellaneous fresh uncured cheeses	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A02QF - Fresh uncured cheese.
A04PP	Other fruit nectars	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A03BB - Fruit nectars (min. 25-50% fruit as defined in EU legislation).
A04PR	Other vegetable juices	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A04PQ - Vegetable juices.
A04PS	Other (mixed) fruit and vegetable juices or nectars	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A0BX9 - Fruit / vegetable juices and nectars.
A04QH	Other liqueurs	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A03NS - Liqueurs.
A04QR	Other regulated additives	Redundant considering the parent term and its applicability according to the FoodEx2 system logic. Children terms previously present under this category are moved one level up i.e. under the term A047R - Food additives other than flavours, colours and artificial sweeteners.
A00TN	Other wild mushrooms	Redundant considering the parent term and its applicability according to the FoodEx2 system logic
A00TV	Other cultivated mushrooms (p)	
A013T	Other pulses (e)	
A017V	Other dried herbs	
A01QL	Other fruit spreads	
A02KH	Other smoked fishes	
A045J	Mixed and other not listed condiments	
A04LS	Other green algae	
A04LT	Other red algae	
A04LV	Other brown algae	
A04MK	Other minor oilseeds	
A04ML	Other minor oilfruit	
A04NE	Other cured meat	

FoodEx2 code	Term name	Reason for deprecation
A04NF	Other cooked cured meat	
A04NQ	Other dairy powder	
A04NR	Other dairy concentrate	
A04NT	Other ice cream and similar	
A04NX	Other processed cheese	
A04PN	Other fruit juices	
A04PT	Other mixed fruit and vegetable juices	
A04QC	Other wine-like fruit drinks	
A04RC	Other leafy vegetables	
A04RL	Other miscellaneous citrus fruit	
A04RY	Other tree nuts (e)	
A04RZ	Other spices seeds (e)	
A04SA	Other spices fruit and berries (e)	
A04SB	Other spices roots or rhizome (e)	
A04SC	Other spices buds (e)	
A04SE	Other stone fruits (e)	

During the catalogue maintenance in 2020, 74 terms were dismissed from the Exposure hierarchy and six terms from the Reporting hierarchy of the FoodEx2 catalogue.

The majority of those dismissed from the Exposure hierarchy (n=66) are related to the sweetening and fortifying agents which should be used only as a facet descriptor to the base terms under the group of 'Table-top sweeteners formulations' and 'Food supplements and similar preparations', respectively. This is to guarantee consistency in reporting of such foods. Therefore, they are no longer available as base terms in the Exposure hierarchy. Moreover, within the major release of next year, in January 2022, they will be dismissed from the Reporting hierarchy as well. Nevertheless, these terms will always be available through the F04-Ingredient and F09-Fortification-agent facets for description of food supplements, table-top sweeteners and fortified foods. Six terms were dismissed due to the alignment with the Commission Regulation (EU) 609/2013 as described in section 3.5.

Additionally, the hierarchy (pyramid) term A01QD - Other processed fruit products (excluding beverages) was set as not reportable in Exposure hierarchy. As described previously in this section, not listed terms should be coded starting from the parent term A01ML – Processed fruit products and added the facet 'F26.A07XE – GENERIC TERM = Other' if facet F27 – Source -commodities cannot be added for more precision. Unlike the other redundant terms that are deprecated as described above, this term is only set as not reportable since it has children terms that remained reportable in different hierarchies and facets.

The term A04RS - Marine fishes not identified was dismissed from the Exposure hierarchy for the same reason as described above. It is still reportable in other hierarchies and facets, to allow data providers

and other users to adjust their coding tables and implement this change in their data sets. However, this term will be deprecated in the major release that will take place in January 2022.

Six terms of different environmental matrices were dismissed from the Reporting hierarchy. Moreover, their type of term and level of detail are changed as they remained reportable only in the MTX hierarchy and in the F02-Part-nature-facet (see section 3.2.). They should be used only as descriptors of a basic FoodEx2 code selected from one of the hierarchies.

Table 5: List of dismissed terms in the FoodEx2 Exposure hierarchy during the maintenance in 2020

FoodEx2 Code	FoodEx2 Term name	Hierarchy in which the term was dismissed
A046M	Artificial sweeteners (e.g., aspartame, saccharin)	Exposure
A046N	Saccharin	Exposure
A046P	Aspartame	Exposure
A046Q	Acesulfame k	Exposure
A046R	Sucralose	Exposure
A046S	Cyclamate	Exposure
A046T	Neo-hesperidine	Exposure
A046V	Thaumatococin	Exposure
A046X	Neotame	Exposure
A046Y	Steviol glucoside	Exposure
A046Z	Advantame	Exposure
A0EVG	Vitamins	Exposure
A0EXZ	Vitamin A (retinol, carotenoids)	Exposure
A0EXY	Vitamin B1 (thiamine)	Exposure
A0EXX	Vitamin B2 (riboflavin)	Exposure
A0EXV	Vitamin B3 (niacin, niacinamide)	Exposure
A0EXT	Vitamin B5 (pantothenic acid)	Exposure
A0EXS	Vitamin B6 (pyridoxine, pyridoxamine, pyridoxal)	Exposure
A0EXR	Vitamin B7 (biotin)	Exposure
A0EXQ	Vitamin B9 (folic acid, folinic acid)	Exposure
A0EXP	Vitamin B12 (cyanocobalamin, hydroxocobalamin, methylcobalamin)	Exposure
A0EXN	Vitamin C (ascorbic acid)	Exposure
A0EXM	Vitamin D (cholecalciferol, ergocalciferol)	Exposure
A0EXL	Vitamin E (tocopherols, tocotrienols)	Exposure
A0EXK	Vitamin K (phyloquinone, menaquinones)	Exposure
A0EVF	Chemical elements	Exposure
A0EXJ	Potassium	Exposure
A0EXH	Calcium	Exposure
A0EXG	Phosphorus	Exposure
A0EXF	Magnesium	Exposure
A0EXE	Zinc	Exposure
A0EXD	Iron	Exposure
A0EXC	Manganese	Exposure
A0EXB	Copper	Exposure
A0EXA	Iodine	Exposure
A0F3A	Fluorine	Exposure

FoodEx2 Code	FoodEx2 Term name	Hierarchy in which the term was dismissed
A0EVZ	Selenium	Exposure
A0EVY	Molybdenum	Exposure
A0EVX	Bromine	Exposure
A16YP	Chromium	Exposure
A0EVS	Special fatty acids	Exposure
A0EVV	Omega-3 fatty acids	Exposure
A0EVT	Omega-6 fatty acids	Exposure
A0EVR	Dietary fibre	Exposure
A0EVM	Phytochemicals	Exposure
A0EVQ	Phytosterols	Exposure
A0EVP	Polyphenols	Exposure
A0EVN	Carotenoids	Exposure
A0EVL	Algae based fortifying agents (e.g. Spirulina, chlorella)	Exposure
A0EVK	Caffeine	Exposure
A16GT	Taurine	Exposure
A16YQ	Glucosamine	Exposure
A0F4M	Co-factors to metabolism	Exposure
A0F4N	Carnitine	Exposure
A0EVJ	Enzymes for fortification	Exposure
A0F4P	Creatine-creatinine	Exposure
A04PB	Other sweetening ingredients	Exposure
A032Z	Polyols	Exposure
A033D	Maltitol	Exposure
A033E	Lactitol	Exposure
A033F	Erythritol	Exposure
A033H	Other polyols	Exposure
A033G	Isomalt	Exposure
A033B	Mannitol	Exposure
A033C	Xylitol	Exposure
A033A	Sorbitol	Exposure
A03RX	Food for sporting people	Exposure
A03RY	Carbohydrate-rich energy food products for sports people	Exposure
A03RZ	Carbohydrate-electrolyte solutions for sports people	Exposure
A03SA	Protein and protein components for sports people	Exposure
A03SB	Micronutrients supplement for sports people	Exposure
A03SC	Carnitine or creatine-based supplement for sports people	Exposure
A01QD	Other processed fruit products (excluding beverages)	Exposure
A04RS	Marine fishes not identified	Exposure
A0C6Z	Air	Reporting
A0C74	Boot swabs	Reporting
A0C70	Delivery box liner	Reporting

FoodEx2 Code	FoodEx2 Term name	Hierarchy in which the term was dismissed
A0C71	Dust	Reporting
A0C72	Fabric swab	Reporting
A0C73	Hatcher basket liner	Reporting

3.9. Revision of applicability or position of existing terms in specific hierarchies and/or facets

In order to adapt the catalogue according to user and stakeholder needs some of existing terms were added to or removed from some hierarchies and/or facets. Reportability was changed for 82 terms in total i.e. 74 terms were dismissed from Exposure hierarchy and six terms were dismissed from the Reporting hierarchy (see section 3.6). The term A0BD5 – Sorghum white (feed) was made reportable in the facet F27 – Source commodities since it was needed for the update done in the Feed section (section 3.3.1.). The term A14RL - Hemerocallis fulva (L.) L. (as plant) was set as reportable in the F01 – Source facet to enable coding of Day lily consumed/analysed in fresh or dry form, as requested by data provider. Parent term was changed for 196 codes in different hierarchies and facets as a consequence of the revision done in the Birds section and deprecations of the redundant terms in the Group for exposure view (see sections 3.2 and 3.6). All changes were implemented in the MTX 12.0 Release of the FoodEx2 catalogue and therefore available in Appendix B –.

4. Conclusions

In 2020 the FoodEx2 catalogue was updated following requests from internal and external users and identified needs in the following domains: Pesticides residues, Food consumption, Chemical contaminants and Botanicals.

Based on suggestions provided by users and stakeholders and requests for addition of missing terms, the following modifications are included in the 2020 FoodEx2 maintenance:

- New terms (n = 108) are added in different hierarchies and facets
- The facet F31-Animal-age-class is amended according to the SIGMA project needs
- The A0CMJ - Hymenoptera (as animal) section is amended according to the MUST-B project needs
- The Birds section is amended to accommodate AI data collection and to complete taxonomy of existing species
- 40 terms are deprecated while 80 terms are dismissed from different hierarchies and facets as found redundant, duplicated or not in line with specific regulations
- Term names, common names, scientific names, implicit attributes and scope notes, term types and level of detail of some existing terms are changed to better specify the scope of the term or to expand its meaning
- Implicit facets were amended or updated for 87 terms
- The reportability of some terms was extended to further hierarchies and/or facets
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found

The present report documents the update of FoodEx2 performed within two minor releases published in 2020 (MTX 11.1 and MTX 11.2) and the major release published in January 2021 (MTX 12.0).

The complete content of the FoodEx2 system is available as Appendix B – to this document as exported spreadsheet MTX_12.0.xlsx¹⁴.

¹⁴ The MTX_12_0.ecf is available at <https://zenodo.org/record/4495166#.YCFhT-hKjD4> in the DCF_catalogues.zip folder. Name of the file is MTX.xlsx

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Abbreviations

AI	Avian Influenza
EC	European Commission
EFSA	European Food Safety Authority
EU	European Union
FoodEx2	Food Classification and Description System
MTX	FoodEx2 Matrix hierarchy
RPC	Raw primary commodity
TG	Technical group

Appendix A – New terms added during the 2020 maintenance

Appendix B – FoodEx2 catalogue MTX_12.0